

Primary UD, TR-XLPE 5 kV - 46 kV

Aluminum Conductor, TR-XLPE Insulated
Concentric Neutral, LDPE Jacket

Features

Low tension stripping compounds

True Triple extrusion system and closed handling raw materials system, to eliminate any contact with ambient, until extrusion of insulation and shields

Application

Underground primary residential and commercial distribution circuits

May be used in wet or dry locations, installed in underground ducts or direct burial

Standards

ICEA S-94-649:
Standard for concentric neutral rated cables 5 to 46 kV

AEIC CS8:
Specifications for extruded dielectric, shielded power cable, rated 5 to 46 kV

SPECIFICATIONS

Maximum operating voltage:

5 kV - 35 kV, 100% & 133% IL

Maximum conductor operation temperatures:

Wet and dry locations:

- Normal: 90°C
- Emergency: 130°C
- Short Circuit: 250°C

Engineering Information

Conductor: Aluminum alloy 1350-H19, compressed Class B stranding, or unilay-compressed per ASTM B 231, or combination unilay (19 wires) per ASTM B 786 *On Request: Solid Aluminum conductor or Copper conductor and/or strand filled*

Sizes: 8 AWG - 2 000 kcmil. *On request larger sizes*

Conductor Shield: Semi conducting cross-linked polyethylene (XLPE). *On request, super-clean and super-smooth materials*

Insulation: Thermoset tree-retardant crosslinked polyethylene (TR-XLPE)

Insulation Shield: Semi conducting cross-linked polyethylene (XLPE)

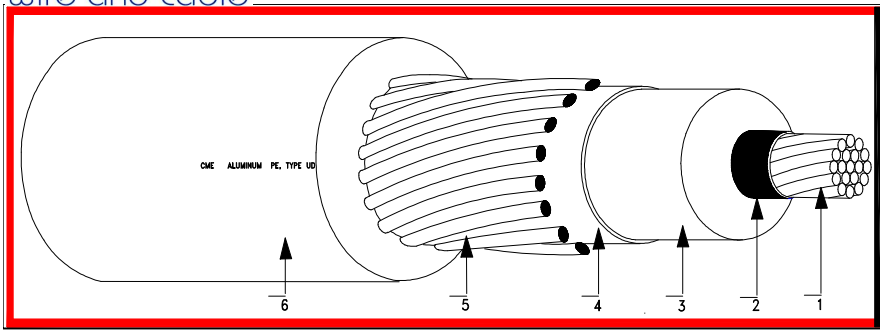
Concentric Neutral: Soft-annealed solid copper wires per ASTM B 3, helically applied and uniformly spaced. Full or 1/3 Neutral. *On request per customer specs,*

Jacket: Extruded to fill (Encapsulated) Black sunlight resistant low density polyethylene (LDPE), with three Red Extruded Stripes. *On request: Overlying (Sleeve) Black polyvinyl chloride (PVC), or semiconducting LDPE*

Configuration Options*: On Request, Triplex or Paralleled configurations*

CME wire and cable

Primary UD, TR-XLPE, 5 kV - 46 kV Aluminum Conductor, TR-XLPE Insulated, Concentric Neutral, LDPE Jacket



1.– ALUMINUM CONDUCTOR COMPRESSED STRANDED
2.– EXTRUDED CONDUCTOR SHIELD
3.– TR–XLPE INSULATION
4.– EXTRUDED INSULATION SHIELD
5.– CONCENTRIC NEUTRAL
6.– ENCAPSULATED POLYETHYLENE JACKET

1/3 NEUTRAL — ***5 kV 100% INSULATION LEVEL***

Size AWG or kcmil	Number of Strands	Insulation Thickness	1/3 Neutral No. of wires	1/3 Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
2	7	90	6	14	50	0.82	332
1	19	90	6	14	50	0.86	363
1/0	19	90	6	14	50	0.90	399
2/0	19	90	7	14	50	0.94	454
3/0	19	90	9	14	50	0.99	530
4/0	19	90	11	14	50	1.05	617
250	37	90	13	14	50	1.10	698
350	37	90	18	14	50	1.21	892
500	37	90	16	12	50	1.39	1225
750	61	90	15	10	80	1.71	1821
1000	61	90	16	9	80	1.88	2318

FULL NEUTRAL — ***5 kV 100% INSULATION LEVEL***

Size AWG or kcmil	Number of Strands	Insulation Thickness	Full Neutral No. of wires	Full Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
2	7	90	10	14	50	0.82	382
1	19	90	13	14	50	0.86	448
1/0	19	90	16	14	50	0.90	519
2/0	19	90	13	12	50	0.98	639
3/0	19	90	16	12	50	1.03	749
4/0	19	90	13	10	50	1.12	933
250	37	90	16	10	50	1.18	1084
350	37	90	16	9	50	1.31	1370

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*5 kV 133% AND 8 kV 100% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	1/3 Neutral No. of wires	1/3 Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
2	7	115	6	14	50	0.87	360
1	19	115	6	14	50	0.91	393
1/0	19	115	6	14	50	0.95	429
2/0	19	115	7	14	50	0.99	486
3/0	19	115	9	14	50	1.04	563
4/0	19	115	11	14	50	1.10	653
250	37	115	13	14	50	1.16	736
350	37	115	18	14	50	1.26	933
500	37	115	16	12	50	1.45	1272
750	61	115	15	10	80	1.76	1878
1000	61	115	16	9	80	1.93	2381

FULL NEUTRAL							*5 kV 133% AND 8 kV 100% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	Full Neutral No. of wires	Full Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
2	7	115	10	14	50	0.871	410
1	19	115	13	14	50	0.91	477
1/0	19	115	16	14	50	0.95	550
2/0	19	115	13	12	50	1.027	672
3/0	19	115	16	12	50	1.077	784
4/0	19	115	13	10	50	1.175	971
250	37	115	16	10	50	1.232	1124
350	37	115	16	9	50	1.359	1414

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*15 kV 100% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**1/3 Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
2	7	175	6	14	50	1.00	435
1	19	175	6	14	50	1.03	470
1/0	19	175	6	14	50	1.07	510
2/0	19	175	7	14	50	1.12	570
3/0	19	175	9	14	50	1.17	652
4/0	19	175	11	14	50	1.22	746
250	37	175	13	14	50	1.28	833
350	37	175	18	14	50	1.41	1064
500	37	175	16	12	50	1.57	1392
750	61	175	15	10	80	1.88	2023
1000	61	175	16	9	80	2.09	2591

FULL NEUTRAL							*15 kV 100% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**Full Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
2	7	175	10	14	50	1.00	485
1	19	175	13	14	50	1.03	555
1/0	19	175	16	14	50	1.07	631
2/0	19	175	13	12	50	1.15	759
3/0	19	175	16	12	50	1.20	875
4/0	19	175	13	10	50	1.30	1070
250	37	175	16	10	50	1.36	1227
350	37	175	16	9	50	1.51	1554

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*15 kV 133% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**1/3 Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
2	7	220	6	14	50	1.09	499
1	19	220	6	14	50	1.13	535
1/0	19	220	6	14	50	1.17	577
2/0	19	220	7	14	50	1.21	640
3/0	19	220	9	14	50	1.26	724
4/0	19	220	11	14	50	1.32	821
250	37	220	13	14	50	1.40	937
350	37	220	18	14	50	1.50	1150
500	37	220	16	12	80	1.73	1558
750	61	220	15	10	80	1.98	2138
1000	61	220	16	9	80	2.18	2719

FULL NEUTRAL							*15 kV 133% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**Full Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
2	7	220	10	14	50	1.09	547
1	19	220	13	14	50	1.13	620
1/0	19	220	16	14	50	1.17	698
2/0	19	220	13	12	50	1.25	831
3/0	19	220	16	12	50	1.30	949
4/0	19	220	13	10	50	1.39	1150
250	37	220	16	10	50	1.48	1337
350	37	220	16	9	50	1.60	1647

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*25 kV 100% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	1/3 Neutral No. of wires	1/3 Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
1	19	260	6	14	50	1.21	597
1/0	19	260	6	14	50	1.25	641
2/0	19	260	7	14	50	1.30	706
3/0	19	260	9	14	50	1.37	818
4/0	19	260	11	14	50	1.43	920
250	37	260	13	14	50	1.48	1014
350	37	260	18	14	50	1.58	1232
500	37	260	16	12	80	1.81	1652
750	61	260	15	10	80	2.09	2296
1000	61	260	16	9	80	2.27	2837
FULL NEUTRAL							*25 kV 100% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	Full Neutral No. of wires	Full Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
1	19	260	13	14	50	1.21	682
1/0	19	260	16	14	50	1.25	762
2/0	19	260	13	12	50	1.33	899
3/0	19	260	16	12	50	1.40	1046
4/0	19	260	13	10	50	1.50	1253
250	37	260	16	10	50	1.56	1418
350	37	260	16	9	80	1.75	1805

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*25 kV 133% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	1/3 Neutral No. of wires	1/3 Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
1	19	320	6	14	50	1.34	699
1/0	19	320	6	14	50	1.40	772
2/0	19	320	7	14	50	1.44	841
3/0	19	320	9	14	50	1.49	933
4/0	19	320	11	14	50	1.55	1038
250	37	320	13	14	50	1.61	1137
350	37	320	18	14	80	1.77	1435
500	37	320	16	12	80	1.94	1801
750	61	320	15	10	80	2.22	2468
1000	61	320	16	9	80	2.39	3022
FULL NEUTRAL							*25 kV 133% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	Full Neutral No. of wires	Full Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
1	19	320	13	14	50	1.34	783
1/0	19	320	16	14	50	1.40	892
2/0	19	320	13	12	50	1.48	1036
3/0	19	320	16	12	50	1.53	1163
4/0	19	320	13	10	50	1.63	1378
250	37	320	16	10	80	1.75	1617
350	37	320	16	9	80	1.87	1949

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*35 kV 100% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**1/3 Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
1/0	19	345	6	14	50	1.45	819
2/0	19	345	7	14	50	1.50	890
3/0	19	345	9	14	50	1.55	983
4/0	19	345	11	14	50	1.60	1090
250	37	345	13	14	80	1.72	1261
350	37	345	18	14	80	1.83	1494
500	37	345	16	12	80	2.02	1916
750	61	345	15	10	80	2.27	2542
1000	61	345	16	9	80	2.44	3102

FULL NEUTRAL							*35 kV 100% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**Full Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
1/0	19	345	16	14	50	1.45	940
2/0	19	345	13	12	50	1.53	1086
3/0	19	345	16	12	50	1.58	1214
4/0	19	345	13	10	80	1.74	1503
250	37	345	16	10	80	1.80	1676
350	37	345	16	9	80	1.93	2012

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL							*35 kV 133% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**1/3 Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
1/0	19	420	6	14	50	1.61	971
2/0	19	420	7	14	50	1.65	1046
3/0	19	420	9	14	80	1.77	1216
4/0	19	420	11	14	80	1.82	1331
250	37	420	13	14	80	1.88	1439
350	37	420	18	14	80	2.02	1734
500	37	420	16	12	80	2.18	2125
750	61	420	15	10	80	2.43	2776
1000	61	420	16	9	80	2.60	3352

FULL NEUTRAL							*35 kV 133% INSULATION LEVEL*
Size AWG or kcmil	**Number of Strands**	**Insulation Thickness**	**Full Neutral**		**Jacket Thickness**	**Approximate Outside Diameter**	**Approximate Net Weight**
			No. of wires	**Size AWG**			
		Mils	**Mils**	**Mils**	**Mils**	**Inches**	**Lbs/Mft**
1/0	19	420	16	14	50	1.608	1092
2/0	19	420	13	12	80	1.75	1317
3/0	19	420	16	12	80	1.8	1452
4/0	19	420	13	10	80	1.899	1684
250	37	420	16	10	80	1.956	1862
350	37	420	16	9	80	2.116	2263

The above data are approximate and subject to normal manufacturing tolerances.

5 kV - 15 kV 100% or 133% Insulation Level Ampacity		
Size AWG or kcmil	† Ampacity	
	Full Neutral	1/3 Neutral
2	125	125
1	145	146
1/0	165	166
2/0	188	189
3/0	213	214
4/0	241	244
250	-	266
350	-	320
500	-	394
750	-	476
1000	-	550

25 kV - 46 kV 100% or 133% Insulation Level Ampacity		
Size AWG or kcmil	† Ampacity	
	Full Neutral	1/3 Neutral
1	147	147
1/0	167	167
2/0	193	194
3/0	218	220
4/0	247	250
250	-	273
350	-	326
500	-	392
750	-	486
1000	-	556

† Ampacities are based on the following:
Three shielded single insulated aluminum conductors in buried ducts, single circuit, 25℃ ambient eart h temperature, 75% load factor, 90 RHO and 90℃ condu ctor temperature (IEEE Std. 835).

Rating Voltage	Minimum Size AWG
8 kV	6
15 kV	2
25 kV	1
35 kV	1/0